



Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaoon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Information Technology

NBA Accredited-UG Programme

Class: T.Y. B. Tech Semester: V

Subject: Foundation of Data Science Laboratory (PEIT317B)

Practical No.: 01

Title:

Assignment based on Numpy arrays and functions (Numpy Library)

Software Requirements:

- Python Recent Version, Visual studio code editor(Any other editor)

Learning Outcomes:

By the end of this assignment, students should be able to:

- Understand the Role of NumPy in Scientific Computing,
- Demonstrate Proficiency in NumPy Array Operations
- Apply Mathematical and Statistical Functions using NumPy
- Solve Real-World Problems Using NumPy
- Bridge Foundational Skills to Advanced Topics

Case Study: Analyzing Student Grades with NumPy, Analyze historical stock prices: Use NumPy to compute moving averages, daily returns, volatility, Use array slicing and vectorized operations for faster analysis etc.

Theory:

- Introduction to NumPy,
- NumPy array creation- `np.array()`, `np.arange()`, `np.linspace()`, `np.linspace` etc.
- NumPy array operation- addition, subtraction, multiplication, division, square, etc.
- NumPy array mathematical operation,
- NumPy array attributes,
- NumPy array indexing,
- `np.arange()`, `np.reshape()`
- self learning Topics (`argmax`, `argmin`, `linspace`, `diff`)

Frequently Asked Questions:

- What is a NumPy array? How is it different from a Python list?
- How do you create a 1D, 2D, and 3D NumPy array?
- How can you create an array of all zeros or all ones?

- What is the use of np.arange() vs np.linspace()?
- How can you check the shape, size, and data type of a NumPy array?
- What does the .ndim attribute return?
- What happens if you use negative indexing in NumPy?
- How do you perform arithmetic operations on NumPy arrays?
- How do you create an array of random integers?

Conclusion:

Thus we have implemented Assignment based on Numpy arrays and functions (Numpy Library)

References:

- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Third Edition ISBN:978-9355421982.
- Spiegel. M.R., Schiller. J. and Srinivasan. R.A., "Schaum's Outline of Theory and Problems of Probability and Statistics", Tata McGraw Hill Edition, 2004.
- Ethem Alpaydin, Introduction to Machine Learning, PHI 4th Edition-2020 ,The MIT Press, ISBN:9780262043793.

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